

Yumaa Hidayat

yumaahidayat@gmail.com | linkedin.com/in/yumaahidayat | github.com/yumaahidayat

EDUCATION

- Tohoku University**, Sendai, Japan Oct. 2022 – Present
Bachelor of Engineering in Mechanical and Aerospace Engineering
- Specialized in Neurorobotics | Current GPA: 2.93/3.00 | Expected Graduation: September 2026
 - Secured the MEXT Scholarship, a four-year fully-funded merit scholarship by the Japanese Government
- SMA Negeri 3 Semarang**, Semarang, Indonesia July 2017 – May 2020
Natural Science Course

RESEARCH & PROJECT EXPERIENCE

- Undergraduate Thesis** Dec. 2025 – Present
A Simulation Study on the Swimming Performance of a Sea Turtle-like Robot
- Developed a physics simulation model in MuJoCo by overhauling multi-body CAD assemblies in Autodesk Inventor, converting them into STL meshes, and defining rigid-body dynamics via MJCF
 - Engineered a bio-inspired gait-phase modulator to alter fin kinematics, outperforming the baseline sinusoidal controller by reducing the cost of transport and increasing swimming velocity at identical frequencies
 - Designed and trained a Reinforcement Learning control framework using Stable-Baselines3, enabling controllable robot with precise velocity and yaw-rate tracking for autonomous path-following across circular, sinusoidal, and lemniscate trajectories
- Design and Drawing II | SolidWorks Final Project** May 2025 – July 2025
Robotic Arm Assembly
- Translated 2D engineering blueprints into 3D CAD models using SolidWorks, implementing custom modifications on individual components to execute motion-conforming robotic arm assembly
 - Conducted kinematic and torque sizing calculations to select standard industrial parts, evaluating gear reduction ratios, interaxial belt lengths, and stepper motors to meet specified operational range constraints
- BIOMOD | Biomolecular Design Competition** June 2024 – Nov. 2024
Bidirectional Nano Grabbers
- Collaborated within a 7-member team to design a biostructure using Actin and modified Thymosin $\beta 4$ for self-assembling string-like fibers
 - Engineered protein sequences via RFDiffusion and performed 3D structural analysis and visualization utilizing AlphaFold3 and PyMOL
 - Validated models against lab experiments, leading the team to secure 4 awards, including 1st Runner Up, Gold Project Award, and 3rd place for Best Website and Video
- Computer Seminar 1 | Pygame Final Project** Apr. 2024 – May 2024
Catapult-based Basketball Game
- Developed a custom 2D physics engine and collision detection system using vector mathematics—including dot products and velocity reflection—to simulate ball trajectories and stable impact responses at 60 FPS
 - Engineered a modular software architecture and state management system with a dynamic difficulty scaling algorithm based on obstacle movement multipliers
 - Conducted a comparative analysis of pixel-to-pixel mask collision versus point-line interaction to eliminate velocity anomalies and optimize runtime stability

ORGANIZATIONAL EXPERIENCE

- External Affairs Staff** at Perhimpunan Pelajar Indonesia Sendai Sep. 2024 – Aug. 2025
- Negotiated a collaboration with the Indonesian Embassy (KBRI Tokyo) to establish a mobile passport renewal service in Sendai to eliminate travel barriers for the local diaspora
 - Initiated a benchmarking initiatives with peer student organizations, facilitating a platform for knowledge transfer
- Event Coordinator** at Perhimpunan Pelajar Indonesia Sendai Sep. 2023 – Aug. 2024
- Coordinated the end-to-end execution of semiannual welcome and graduation parties, managing budgeting, venue logistics, and agenda planning for over 100 attendees per event

SKILLS

Programming Languages: Python, C/C++ (Familiar)
Libraries & Tools: NumPy, Matplotlib, MuJoCo (MJCF), Gymnasium, Stable-Baselines3, TensorBoard, Pygame
Software: SolidWorks, Autodesk Inventor
Spoken Languages: Native Bahasa Indonesia, Fluent English, JLPT N3-level Japanese